Label Semantic Dynamic Guidance Network for Remote Sensing Image Scene Classification

Borui Chai, Tianming Zhao, Runou Yang, Nan Zhang, Tian Tian, and Jinwen Tian

***Abstract*— The remote sensing image scene classification continues to face significant challenges due to high intraclass diversity and interclass similarity. Existing methods mainly use semantic associations between images to establish deep semantic associations between classes, ignoring the rich high-level semantic knowledge contained in the label text. This high-level information are especially valuable for distinguishing between confusing categories, as it enables the model to capture both similarity and distinctive features effectively. In this article, we introduce a novel approach that incorporates label semantic information and proposes a plug-and-play framework to guide classification model learning of intraclass and interclass relationships. Specifically, our framework includes a dynamic soft label module (DSLM), which uses textual semantics to facilitate classification model learning of interclass relationships via soft labels at the target level. In addition, we design a coarse-to-fine contrastive module (CFCM) to integrate textual semantics into contrastive learning, guiding the model in capturing intraclass and interclass relationships at the feature level. Our framework is compatible with both convolutional neural network (CNN)-based and vision transformer (ViT)-based classification architectures and is employed solely during training to minimize computational overhead. Experimental results on four datasets validate the effectiveness of our approach.**

***Index Terms*— Contrastive learning, dynamic soft label generation, label semantic information, label semantics, remote sensing image scene classification.**

I. Introduction

With the advancement of spaceborne remote sensing technology, it has become increasingly easy to obtain high-resolution remote sensing images by observing the Earth through satellites [1]. The vast amount of spaceborne remote sensing images plays a crucial role in various fields, such as urban and regional planning and natural resources conservation and management [2]. The remote sensing scene classification is a fundamental task in the application of remote sensing images and serves as a key component in the broader utilization of remote sensing technology. The remote sensing image scene classification faces two major challenges: strong intraclass diversity and inevitable interclass similarity.

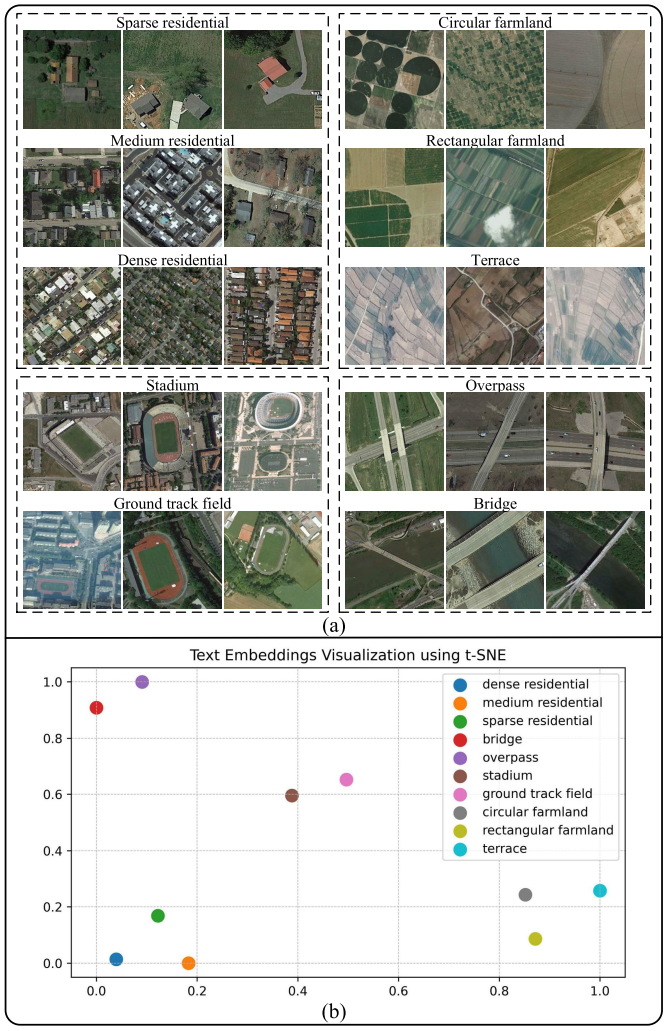


Fig. 1. Categories with high interclass similarity also have similarities in the semantic information of their labels. (a) Interclass similarity of scenes. (b) Visualization results of category text vectors by t-SNE dimensionality reduction.

Specifically, the intraclass diversity is manifested as different image features of the same type of ground area under different climatic conditions, geographical environments, architectural styles, and shooting distances. The interclass similarity is manifested by the presence of similar feature regions and high semantic overlap in different categories. As shown in Fig. 1(a), residential areas of different densities all contain the feature of houses. Therefore, further improving the performance of remote sensing image scene classification under complex semantic overlap remains an important and challenging goal.

In recent years, deep neural networks (DNNs) have made significant progress in the field of computer vision. DNNs

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The authors are with the National Key Laboratory of Multispectral Information Intelligent Processing Technology, School of Artificial Intelligence and Automation, Huazhong University of Science and Technology, Wuhan 430074, China (e-mail: br_chai@hust.edu.cn; tming@hust.edu.cn; runouyang@hust.edu.cn; zhang_nan@hust.edu.cn; ttian@hust.edu.cn; jwtian@mail.hust.edu.cn).

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have powerful representation capabilities and have brought significant performance improvements to most computer vision tasks. In remote sensing image scene classification tasks, most DNN-based methods focus on modifying the network structure, aiming to improve the model's ability to extract features from images. Although these methods have achieved good results, they ignore the semantic connections between images, which limits their performance in easily confused fine-grained categories. There are already some methods attempting to explore deep semantic correlations between remote sensing scenes. Generally speaking, these methods can be divided into two categories: target-level semantic association mining [3], [4], [5], [6], [7], [8] and feature-level semantic association mining [9], [10], [11], [12]. The method based on target-level semantic association mining constructs soft labels through knowledge distillation or label distribution learning. The method based on feature-level semantic association mining optimizes the network by optimizing the intraclass and interclass distances, intuitively obtaining semantic relationship information between classes. However, the two types of methods described above are essentially still based on learning the semantic correlation of categories from images, ignoring that inherent in the category text itself.

To address the above limitations, we would like to introduce label semantic information that simultaneously guides the network to learn correlations between categories at both the feature level and the target level, leading to more accurate classification. The label text itself is a description of the overall features of the image, and the links between labels are unified with the links between images. To further exploit this unity, we utilize the large language model (LLM) to generate knowledgeable textual descriptions for scene labels that contain rich human high-level semantic information. As shown in Fig. 1(b), we generate the description information of the label text by LLM, and then generate the label text vectors by text encoder and visualize the display using t-distributed stochastic neighbor embedding (t-SNE) [13]. From the figure, it can be seen that images with high interclass similarity have relatively small distances between their label text vectors. Therefore, the image classification model can learn the connection between classes guided by the semantic information of the labels.

Based on the above analysis, this article proposes a label semantic dynamic guidance network (LSDGNet), which is a new architecture that uses label semantic information to dynamically guide the learning of remote sensing scene classification models at the target and feature levels. A brief schematic of the structure of LSDGNet is shown in Fig. 2. During the training phase, we use text semantics to guide the network to implicitly learn intraclass and interclass relationships. We design a dual-branch architecture, which divides the entire framework into a remote sensing scene classification branch and a cross-modal guidance branch. The remote sensing scene classification branch is used for image feature extraction and classification, while the cross-modal guidance branch is responsible for guiding the learning of remote sensing scene classification. In the cross-modal guidance branch, we input the label semantic information generated by LLM into a frozen text encoder to obtain the initial

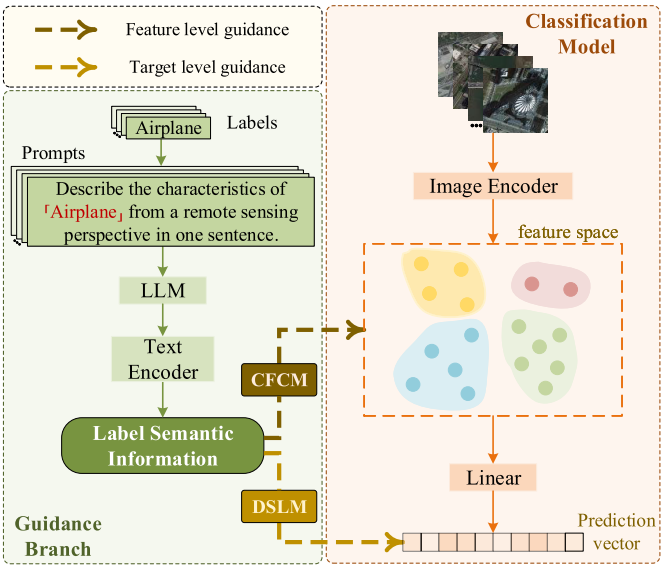


Fig. 2. Brief structural diagram of LSDGNet. Label semantic information are used to guide model learning uniformly at the feature level and the target level. DSLM and CFCM guide the classification model to learn connections between categories at the target level and feature level, respectively. A detailed explanation of the two modules can be found in Section III.

category semantic representation vector, which is dynamically updated through a designed linear layer. Then, we use a hierarchical clustering algorithm to cluster the semantic representation vectors of categories into multiple coarse-grained categories. Labels of the same coarse-grained category have high semantic correlation. Based on the results of semantic dynamic clustering, we design a dynamic soft label module (DSLM) and a coarse-to-fine contrastive module (CFCM). The DSLM restricts softened labels to the same coarse-grained categories, reducing the introduction of nonrelevant category noise. The CFCM gradually optimizes the learning process by decoupling the contrastive loss, guiding the network to learn the hierarchical associations between categories.

The main contributions of this article are summarized as follows.

1) We propose LSDGNet, which uses category semantic information to guide the model to gradually mine category associations for accurate remote sensing scene classification.
2) We design a DSLM and a CFCM to guide the model to learn semantic correlations in remote sensing scenes at both the target and feature levels.
3) We propose a plug-and-play dual-branch architecture. During the reasoning process, the text-guided branch can be removed without adding any additional burden.
4) We conduct comprehensive experiments on four popular public datasets to demonstrate the effectiveness of our approach.

The remainder of this article is organized as follows. Section II reviews the related works on remote sensing image scene classification, interclass semantic association mining, and visual language models (VLMs) in remote sensing scene classification. Section III details the proposed LSDGNet framework and its key components. In Section IV, we present detailed experimental results on four datasets and analyze the

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results. Finally, Section V summarizes the main work of this article.

II. Related Works

A. Remote Sensing Image Scene Classification

Early feature-based methods are mostly handcrafted by domain experts to extract fixed image features. Techniques such as scale-invariant feature transform (SIFT) [14], local binary pattern (LBP) [15], gray-level co-occurrence matrix (GLCM) [16], and histogram of oriented gradients (HOGs) [17] are designed to capture low-level image features such as texture, color, gradients, and shape. However, these methods struggle to comprehensively represent the rich semantic information in remote sensing images under complex conditions, which limits their performance in remote sensing scene classification.

To further improve the utilization and quantification of handcrafted image features, intermediate feature extraction methods are proposed. The bag of visual words (BoVWs) model [18] treats extracted image features as visual words and represents the image as a feature vector using statistical information for classification. Although intermediate features capture more complex and deeper characteristics, they still rely on handcrafted features and struggle to represent deep features effectively.

In recent years, with the advent of DNNs for learning powerful feature representations, the performance of remote sensing image scene classification has significantly improved. Many studies have focused on enhancing the feature extraction capabilities of DNNs. ResNet [19], EfficientNet [20], and vision transformer (ViT) [21] are among the most commonly used baseline models today. Scholars have also aimed to improve the extraction of important discriminative features, such as local discriminative features and global contextual information. For example, Zhang et al. [22] feed convolutional neural network (CNN)-extracted features into CapsNet to strengthen the hierarchical spatial feature encoding. Lu et al. [23] propose an end-to-end feature aggregation CNN, exploring feature aggregation in intermediate network layers. Zhao et al. [24] combine global background and local discriminative features in a unified framework called co-enhanced global-part integration network (CGINet), utilizing a lightweight context-aware attention module to capture a larger receptive field and contextual information. Lv et al. [25] propose spatial-channel feature preserving ViT (SCViT), which not only considers local information between patches in ViT but also integrated global information in both the spatial and channel domains, significantly enhancing ViT's performance. The above methods focus on improving the network's ability to extract features from images, ignoring the semantic information association between different categories, which is particularly important in categories that are prone to confusion.

B. Interclass Semantic Association Mining

We summarize the methods based on interclass semantic association mining into two categories: target-level semantic association mining and feature-level semantic association mining.

The semantic association mining between target-level classes provides additional supervision for the model by designing informative soft labels. The existing methods mainly include manually designed supervised information, label distribution learning, and knowledge distillation. Xie et al. [5] add additional rotation angle information to the labels for supervision while performing rotation enhancement on the images. However, the information brought by artificially defined additional labels are limited. Luo et al. [6] construct a neighbor-based label distribution learning task by jointly capturing local similarity and label correlation, and use the learned labels for training, enriching the interclass association information. However, the label distribution cannot be dynamically updated during training. Wang et al. [3] generate adaptive soft labels to assist in training based on the statistical data of each category predicted by the classification model, but only use the final prediction results as the information contained in the soft labels, which is relatively limited. Hu et al. [7] achieve hierarchical extraction of interclass relationships by constructing class entanglement information using feature prediction vectors of models at different scales. He et al. [8] propose correlation label embedding network (CLENet) to construct soft labels using predictive information from the network. These methods are still based on image modeling of interclass relationships, ignoring the semantic associations inherent in categorical texts.

Feature-level semantic association mining guides models to increase interclass distance and decrease intraclass distance through metric learning or contrastive learning. For example, Cheng et al. [9] add metric learning regularization to convolutional neural networks for feature extraction, adjusting the distance between feature vectors. Wang et al. [12] combine triplet metric with multiscale attention module to optimize feature learning at different scales. However, due to the richer semantic information in remote sensing scenes, directly applying metric learning is not optimal. Bi et al. [11] introduce supervised contrastive learning in ViT and use a two-stage training approach of cross-entropy loss contrastive loss to improve performance. Zhao et al. [10] propose a multisample contrastive network, which introduces a feature dictionary to expand the number of contrastive samples in supervised contrastive learning. The above methods have achieved excellent results, but they did not consider the hierarchical relationship between classes in the contrastive process.

This article utilizes label semantic information to construct hierarchical relationships, unifying target-level and feature-level semantic mining methods into a unified framework.

C. VLMs for Remote Sensing

With the rapid development of VLMs and LLMs, utilizing existing multimodal models to assist in solving domain-specific downstream tasks has become an area of active research [26]. Qiu et al. [27] propose a prototype-based method for remote sensing image scene classification that requires only a small number of labeled samples, leveraging

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open self-supervised features from cross-modal feature-aligned contrast learning (CLIP) without additional training. However, the accuracy achieved by this method remains limited. An effective approach to applying VLMs to downstream remote sensing tasks is fine-tuning. For instance, Zhang et al. [28] introduce GeoRSCLIP by proposing the large-scale remote sensing image–text pairing dataset, RS5M, and using it to fine-tune the CLIP model. Liu et al. [29] propose RemoteCLIP by integrating unmanned aerial vehicle (UAV) images to construct a pretraining dataset. However, these methods rely on large-scale data, leading to significant training overhead. Cai et al. [30] employ a VLM to generate textual descriptions for images and fuse both image and text features to enhance classification performance. However, this approach overlooks the semantic attributes inherent in image labels, introducing additional training and inference burdens.

In this article, we utilize an LLM and a VLM to extract the semantic information embedded in label text and use it to guide the classification model in learning both intraclass and interclass relationships.

III. Method

The framework of LSDGNet is displayed in Fig. 3. LSDGNet contains two branches: the remote sensing scene classification branch and the cross-modal guidance branch. The remote sensing scene classification branch takes remote sensing images as input, encodes them and finally performs scene classification. We use the proposed cross-modal guidance branch to guide the classification branch in learning the image features. Specifically, we allow LLM to generate label descriptions through designed prompts and generate label text vectors through a frozen text encoder. We obtain the interclass similarity matrix by computing the similarity between label text vectors and dynamically update the interclass similarity matrix with a fine-tunable linear layer and exponential moving average (EMA) strategy. At the same time, we perform hierarchical clustering on the label text vectors to construct coarse-grained categories. Based on the coarse-grained categories we design a DSLM and a CFCM to guide the learning of the remote sensing scene classification branch in both target and feature dimensions. In the DSLM, we construct soft labels for categories based on the interclass similarity matrix and restrict the softened categories to coarse-grained categories to remove irrelevant category noise. In the CFCM, we decouple the contrastive loss and gradually optimize the distance between coarse-grained categories and fine-grained categories to construct interclass hierarchical relationship. Compared with previous classification models constructed using CLIP, our approach does not directly align image and text features. This strategy reduces the model's reliance on large-scale data and training resources. Furthermore, in our method, the cross-modal guidance branch can be removed during the inference stage, significantly reducing the additional complexity introduced during inference.

A. Label Dynamic Clustering

1) Label Text Embedding: Our motivation is to leverage the high-level human knowledge embedded in LLM to guide the network in learning the relationships between different categories, thereby achieving hierarchical and high-quality rapid learning. Label text represents a type of multimodal information that is easily accessible and contains advanced human knowledge. LLM can further elaborate on the specific meanings of these text labels, generating corresponding textual descriptions. By designing prompts specific to remote sensing scenarios, we can extract detailed textual descriptions for each category, which contain high-level semantic information about the remote sensing scene. We then use a frozen text encoder to convert these textual descriptions into corresponding text vectors. The above process can be formulated as follows:

$$\mathbf{v_i} = \text{TextEncoder}(\text{LLM}(\text{prompt}_i)) \tag{1}$$

where $\mathbf{v_i}$ is the category text vector corresponding to the ith category.

2) Dynamic Clustering: In an image classification task, for a dataset containing n categories, a text vector matrix $M_t = [\mathbf{v_1}, \mathbf{v_2}, \mathbf{v_3}, \ldots, \mathbf{v}_n]$ can be obtained through the text encoder. In the overall optimization process, there is an implicit alignment process between the label text features in the bootstrap branch and the image features in the classification branch, which requires fine-tuning the parameters of the text encoder. However, fine-tuning the entire text encoder imposes a significant training burden. To address this problem, we design a linear layer to learn and fine-tune the feature attributes of the text vectors

$$M_t' = \text{Linear}(M_t) \tag{2}$$

where M_t' represents the fine-tuned text vector matrix. We initialize the linear layer as an identity matrix and apply strong regularization to ensure that the original feature attributes do not drift excessively. In each minibatch, the label text vectors are updated through the linear layer, and a label similarity matrix is computed for all label text vectors. The above process can be formalized as follows:

$$M_{\text{similarity}} = M_t' \cdot M_t'^{\text{T}} = [\mathbf{s_1}, \mathbf{s_2}, \mathbf{s_3}, \ldots, \mathbf{s}_n] \tag{3}$$

where $\mathbf{s_i} = [\sigma_1, \sigma_2, \sigma_3, \ldots, \sigma_n]$ represents the similarity between each category text vector and other category text vectors. Based on the category text similarity matrix and the similarity threshold k, we use hierarchical clustering to divide multiple fine-grained categories into multiple coarse-grained categories

$$C = [\mathbf{c_1}, \mathbf{c_2}, \mathbf{c_3}, \ldots, \mathbf{c}_n] = \text{Cluster}(M_{\text{similarity}}, k) \tag{4}$$

where C is the coarse category mask matrix, where each vector $\mathbf{c_i} = [\theta_1, \theta_2, \theta_3, \ldots, \theta_n]$ corresponds to the mask vector for each category, indicating the coarse category to which the given category belongs.

B. Dynamic Soft Label Generation

We expect to use the label semantic information to dynamically guide the model to mine interclass relationship information during model training. Thus, guidance at the target level requires dynamically changing soft labels. The DSLM is shown in Fig. 4. We generate the category similarity matrix $M_{\text{similarity}}$ using the text vector matrix M_t' according to (3). For

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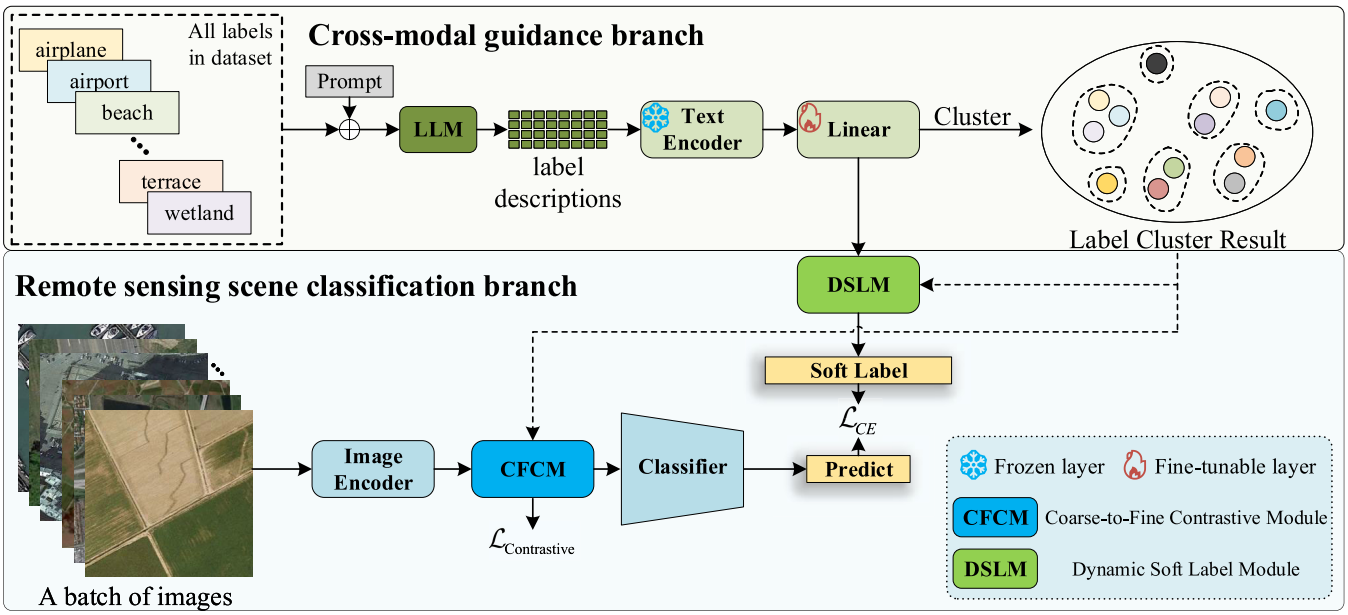


Fig. 3. Overall model architecture. We design a cross-modal guidance branch to dynamically guide the network's learning of image features. The inference process uses only the remote sensing image scene classification branch without any textual inputs.

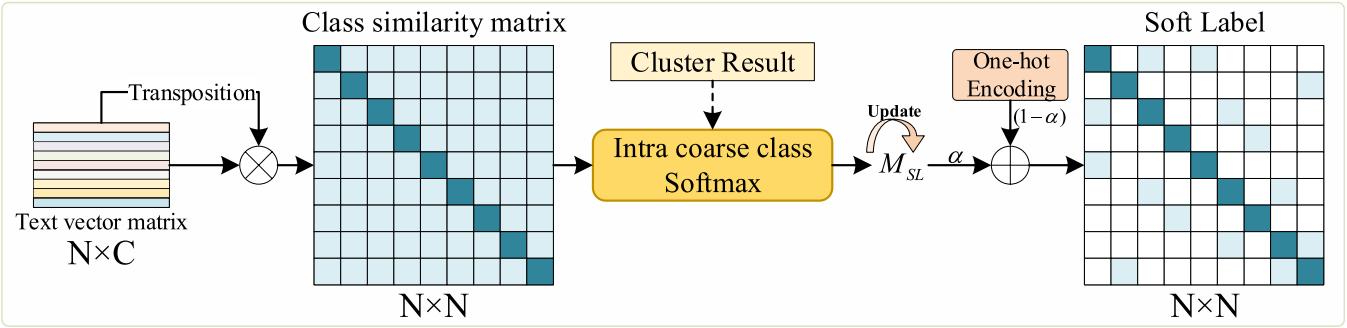


Fig. 4. Architecture of DSLM. M_{SL} is the transition soft label matrix, which we initialize with the original text vector matrix, and then update it continuously with EMA. $\otimes$: matrix multiplication and $\oplus$: matrix addition.

the association vector $\mathbf{s_i}$ of an individual category in matrix $M_{\mathrm{similarity}}$, we extract the similarities between that category and the other categories within the same coarse category

$$\mathbf{s_i^{selected}} = [\sigma_j | \theta_j = 1 \quad \forall j \in \{1, 2, \ldots, n\}] \tag{5}$$

$$\mathrm{Indices}_i = \{j | \theta_j = 1, j \in \{1, 2, \ldots, n\}\} \tag{6}$$

where $\mathbf{s_i^{selected}}$ represents the extracted similarity vector that contains only the similarities between the current category and those within the same coarse-grained category, including itself. $\mathrm{Indices}_i$ refers to the index positions of the elements in $\mathbf{s_i}$. These indices are used to construct the dynamic soft labels. These similarities are normalized using a softmax function with temperature τ

$$\mathbf{p}_i{}^{\mathrm{selected}} = \mathrm{softmax}\left(\mathbf{s_i^{selected}}/\tau\right). \tag{7}$$

The normalized values are placed back into a zero-initialized vector $\mathbf{p}_i$, which has the same dimensions as $\mathbf{s_i}$, based on the indices $\mathrm{Indices}_i$

$$\mathbf{p}_i = \mathrm{scatter}\left(\mathbf{p}_i{}^{\mathrm{selected}}, \mathrm{Indices}_i\right) \tag{8}$$

where the scatter operation is to fill $\mathbf{p}_i{}^{\mathrm{selected}}$ into $\mathbf{p}_i$ according to $\mathrm{Indices}_i$. The transition soft label matrix M_{SL} consists of $\mathbf{p}_i$. We initialize M_{SL} with the original text vector matrix M_t and update it based on EMA [31]. ω is set to 0.9995

$$M_{\mathrm{SL}}^{(k)} = \omega M_{\mathrm{SL}}^{(k-1)} + (1-\omega)\Delta M_{\mathrm{SL}}. \tag{9}$$

Soft labels provide interclass relationship information, while hard labels provide clear classification boundaries. To fully leverage the advantages of both soft and hard labels, the training process should be guided by soft labels in the early and middle stages to provide the model with abundant auxiliary information, while hard labels should dominate in the later stages to further emphasize the differences between categories. Therefore, we use the following equation for dynamic label fusion:

$$\widetilde{\mathbf{y}}_i = \alpha \mathbf{p}_i + (1-\alpha)\mathbf{y}_i \tag{10}$$

where $\mathbf{y}_i$ represents the one-hot encoded hard label for category i. The weighting parameter α is reduced in iterations by a cosine decay strategy [32], thus contributing to enhanced

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convergence. α is computed using the following equation:

$$\alpha = \frac{1}{2}\left(1 + \cos\left(\frac{e}{e_0}\pi\right)\right) \tag{11}$$

where e is the number of epochs and e_0 represents the aggregate count of epochs.

C. Coarse-to-Fine Contrastive Learning

Most current contrast learning methods ignore the shared attributes between categories, resulting in models that ignore associations between different categories. Thanks to the association property inherent in the semantic information of labels, based on the work of [33], we introduce a decoupled contrast loss that hierarchically optimizes the distance between features. Specifically, based on the coarse-grained categories generated by the cross-modal guidance module, we design a coarse-to-fine decoupled contrastive module. As shown in Fig. 5, we first minimize the similarity between all coarse categories at the coarse-grained level, and then further minimize the similarity between different fine-grained categories within the same coarse category at the fine-grained level.

Let $\mathcal{N}$ and $\mathcal{P}$ be the sets of negative and positive sample pairs, respectively. In the coarse-grained contrastive process, positive samples are defined as pairs of samples belonging to the same coarse category, while negative samples are defined as pairs of samples belonging to different coarse categories. In the fine-grained contrastive process, positive samples are defined as pairs of samples that belong to both the same coarse and fine categories, whereas negative samples are defined as pairs of samples that belong to the same coarse category but different fine categories. Formally, the set of positive and negative sample pairs in the coarse-grained contrastive process and the fine-grained contrastive process can be represented as follows:

$$\mathcal{P}_{\text{coarse}} = \left\{(i, j) \middle| y_i^c = y_j^c\right\} \tag{12}$$

$$\mathcal{N}_{\text{coarse}} = \left\{(i, j) \middle| y_i^c \neq y_j^c\right\} \tag{13}$$

$$\mathcal{P}_{\text{fine}} = \left\{(i, j) \middle| y_i^c = y_j^c \text{ and } y_i^f = y_j^f\right\} \tag{14}$$

$$\mathcal{N}_{\text{fine}} = \left\{(i, j) \middle| y_i^c = y_j^c \text{ and } y_i^f \neq y_j^f\right\} \tag{15}$$

where y_i^c represents the coarse-grained category to which the ith sample belongs. y_i^f represents the fine-grained category to which the ith sample belongs, i.e., the original category of the sample. To achieve minimizing interclass similarity and maximizing intraclass similarity, we define the optimization objective function as follows:

$$\begin{aligned}\mathcal{L} &= \log\left(1 + \sum_{i\in\mathcal{P}}\sum_{j\in\mathcal{N}} \exp\left(\text{sim}_n^j - \text{sim}_p^i\right)\right) \\ &= \log\left(1 + \sum_{j\in\mathcal{N}} \exp\left(\text{sim}_n^j\right) \sum_{i\in\mathcal{P}} \exp\left(-\text{sim}_p^i\right)\right)\end{aligned} \tag{16}$$

where sim_n^j represents the cosine similarity between the jth negative sample pair and sim_p^i represents the cosine similarity between the ith positive sample pair. The gradient optimization of the positive and negative sample pairs is controlled using the parameters γ_p^i and γ_n^j, which are determined by the difference between the positive and negative sample pair similarity and the positive and negative best similarity values $\mathcal{D}_p$ and $\mathcal{D}_n$, respectively,

$$\gamma_n^j = \max\left(0, \text{sim}_n^j - \mathcal{D}_n\right) \tag{17}$$

$$\gamma_p^i = \max\left(0, \mathcal{D}_p - \text{sim}_p^i\right). \tag{18}$$

Thresholds m_p and m_n for positive and negative samples are introduced. The final coarse and fine grain losses can be expressed as follows:

$$\begin{aligned}\mathcal{L}_{\text{Coarse}} = \log\Bigg(1 + \sum_{j\in\mathcal{N}_{\text{coarse}}} \exp\left(\gamma_n^j\left(\text{sim}_n^j - m_n\right)\right) \\ \times \sum_{i\in\mathcal{P}_{\text{coarse}}} \exp\left(-\gamma_p^i\left(\text{sim}_p^i - m_p\right)\right)\Bigg)\end{aligned} \tag{19}$$

$$\begin{aligned}\mathcal{L}_{\text{Fine}} = \log\Bigg(1 + \sum_{j\in\mathcal{N}_{\text{fine}}} \exp\left(\gamma_n^j\left(\text{sim}_n^j - m_n\right)\right) \\ \times \sum_{i\in\mathcal{P}_{\text{fine}}} \exp\left(-\gamma_p^i\left(\text{sim}_p^i - m_p\right)\right)\Bigg).\end{aligned} \tag{20}$$

Similar to the process of generating soft labels, we introduce a weighting coefficient α to fuse the decoupled contrastive losses. This coefficient is calculated by (11). The overall contrastive loss can be expressed as follows:

$$\mathcal{L}_{\text{Contrastive}} = (1 - \alpha) \cdot \mathcal{L}_{\text{Coarse}} + \alpha \cdot \mathcal{L}_{\text{Fine}}. \tag{21}$$

In summary, our model is trained with the sum of cross-entropy loss $\mathcal{L}_{\text{CE}}$ and contrastive $\mathcal{L}_{\text{Contrastive}}$ together which can be expressed as follows:

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{CE}}(\mathbf{y}_{\text{pred}}, \widetilde{\mathbf{y}}) + \mathcal{L}_{\text{Contrastive}} \tag{22}$$

where $\mathcal{L}_{\text{CE}}(\mathbf{y}_{\text{pred}}, \widetilde{\mathbf{y}})$ is the cross-entropy loss between the predicted label $\mathbf{y}_{\text{pred}}$ and the soft label $\widetilde{\mathbf{y}}$.

IV. Experiment and Analysis

A. Datasets Description

In order to verify the validity of LSDGNet, four datasets are selected: the UCM dataset [34], the WHU-RS19 dataset [35], the AID dataset [36], and the NWPU-RESISC45 (NWPU) dataset [37].

1) UCM Dataset: The UCM dataset contains 21 categories of scenes, 100 images per category, for a total of 2100 images. The spatial resolution of this dataset is about 0.3 m, and the image scale is 256 pixels × 256 pixels. The images are all extracted from USGS National Map.

2) WHU-RS19 Dataset: The WHU-RS19 dataset contains 19 categories of scenes, each with around 50 images, for a total of 1005 images. The source of the dataset is Google satellite imagery, with an image size of 600 pixels × 600 pixels.

3) AID Dataset: The AID dataset contains 30 classes of scenes, each with about 200–420 images, totaling 10 000 images. The spatial resolution of this dataset ranges from 0.5 to 8 m and the image size is 600 pixels × 600 pixels. The images are obtained from Google Earth.

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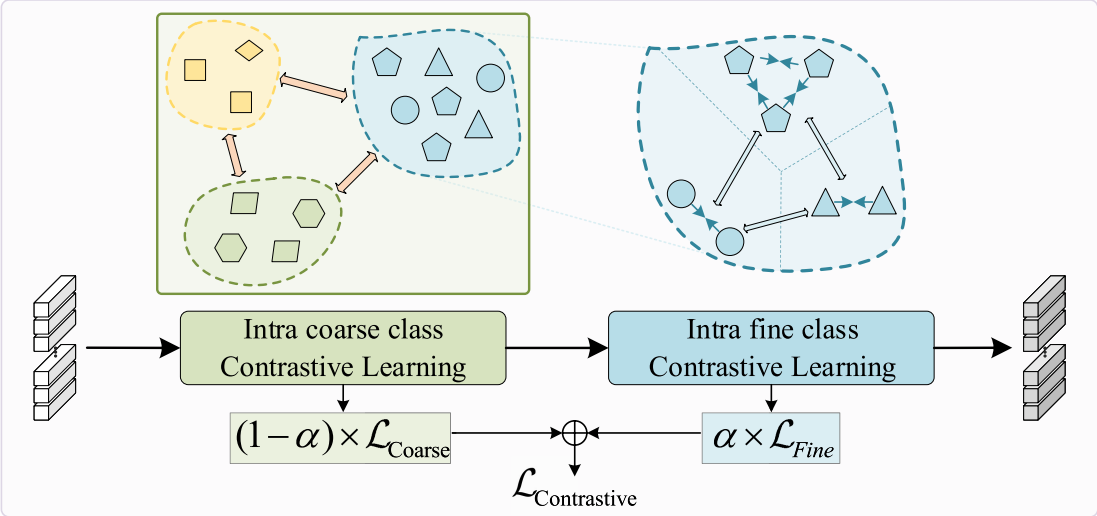


Fig. 5. Architecture of CFCM. The contrastive loss is decoupled into $\mathcal{L}_{\text{Coarse}}$ and $\mathcal{L}_{\text{Fine}}$ by redefining the set of positive and negative samples, and then, the decoupled loss is reweighted to get $\mathcal{L}_{\text{Contrastive}}$.

4) NWPU Dataset: The NWPU dataset covers 45 scene classes with 700 images in each class, totaling 31 500. The spatial resolution of this dataset ranges from 0.2 to 30 m, and the image size is 256 pixels × 256 pixels. This dataset is one of the largest remote sensing scene classification datasets available.

B. Implementation Details

For all experiments, we use the pytorch framework on a GeForce RTX 3090. We select ImageNet [38] pretrained ViT-B and EfficientNet-B3 as the backbone of our remote sensing scene classification branch, respectively. For LLM we choose GPT-4o [39] and for text encoder we choose the text encoder in RemoteCLIP [29]. Random cropping and random flipping are used for data enhancement during training. All input images are resized to 256 × 256 pixels. We train 100 epochs with a batch size set to 16. We train the remote sensing scene classification branch using SGD optimizer with momentum set to 0.9, and weight decay set to 0.0005. For EfficientNet-B3, we set the learning rate to 0.01, and for ViT-B, we set the learning rate to 0.001. The fine-tunable linear layer in the cross-modal bootstrap branch was trained using the SGD optimizer with momentum set to 0.9, weight decay set to 0.5, and learning rate initialized to 0.001. The learning rates are all based on a cosine decay strategy.

All images in the dataset are randomly assigned to either the training set or the test set. Specifically, the proportion of training data in UCM is 50% and 80%, respectively. The proportion of training data in WHU-RS19 is 20% and 50%, respectively. The proportion of training data in AID is 20% and 50%, respectively. The proportion of training data in NWPU is 10% and 20%, respectively. All experiments are repeated five times and the mean overall accuracy (OA) and standard deviation (STD) are reported to avoid any bias caused by random sampling. In addition, we plot a confusion matrix reflecting the extent of classification errors and interclass confusions.

TABLE I

EXPERIMENTAL RESULTS OF DIFFERENT METHODS ON UCM DATASET. (†: CNN-BASED METHODS. ‡: VIT-BASED METHODS. *: HYBRID ARCHITECTURE-BASED METHODS)

Method	OA±STD(%)	
	50% Training	80% Training
GoogLeNet [40] †	92.70±0.60	94.31±0.89
VGGNet16 [41] †	94.14±0.69	95.21±1.20
D-CNN [9] †	-	98.93±0.10
ResNet50 [19] †	98.30±0.20	98.95±0.80
CSCANet [42] †	-	99.76±0.16
EfficientNetB3 [43] †	98.48±0.18	99.33±0.53
MGSNet [44] †	-	99.76±0.14
CGINet [24] †	-	**99.84±0.16**
ViT-B [21] ‡	97.30±0.32	98.81±0.43
V16_21k [45] ‡	98.14±0.47	-
MiniViT [46] ‡	98.88±0.18	99.10±0.25
SCViT [25] ‡	98.90±0.19	99.57±0.31
ET-GSNet [4] ‡	-	99.29±0.34
EMTCAL [43] *	98.67±0.16	99.57±0.28
LSDGNet(ViT-B) ‡	98.55±0.15	99.10±0.35
LSDGNet(EfficientNetB3) †	**99.03±0.07**	99.57±0.27

C. Comparative Study

In order to validate the effectiveness of LSDGNet, some current mainstream algorithms are used to compare the four public benchmark datasets. We classify the methods used for comparison into three categories: methods based on CNN, methods based on ViT, and methods based on hybrid architectures. The detailed analysis of each dataset is as follows.

1) UCM Dataset: The results on the UCM dataset are shown in Table I. When using 50% and 80% of the training samples, respectively, LSDGNet (ViT-B) achieves an absolute improvement of 1.25% and 0.29% compared with the original baseline. LSDGNet (EfficientNetB3) achieves absolute improvements of 0.55% and 0.24%, respectively. This demonstrates the effectiveness of our method on small-scale datasets. Fig. 6(a) and (b) shows the confusion matrices of

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TABLE II
EXPERIMENTAL RESULTS OF DIFFERENT METHODS ON WHU-RS19 DATASET. (†: CNN-BASED METHODS. ‡: VIT-BASED METHODS. *: HYBRID ARCHITECTURE-BASED METHODS)

Method	OA±STD(%)	
	20% Training	50% Training
GoogLeNet [40] †	78.41±0.42	90.57±1.11
VGGNet16 [41] †	93.05±1.05	97.39±0.61
ResNet50 [19] †	95.50±1.06	98.41±0.37
DenseNet121 [47] †	96.29±1.01	98.42±0.48
EfficientNet-B3 [43] †	94.85±0.86	98.46±0.34
ViT-B [21] ‡	94.06±0.75	97.38±0.49
VisFormer [48] ‡	92.23±1.66	96.87±0.58
DeiT [49] ‡	96.88±0.57	98.89±0.32
EMTCAL [43] *	-	97.60±0.23
SF-MSFormer [50] *	96.63±0.23	97.80±0.39
LGLFormer [51] *	96.62	99.20±0.06
LSDGNet(ViT-B) ‡	94.96±0.57	98.22±0.33
LSDGNet(EfficientNet-B3) †	**97.38±0.69**	**99.29±0.29**

TABLE III
EXPERIMENTAL RESULTS OF DIFFERENT METHODS ON AID DATASET. (†: CNN-BASED METHODS. ‡: VIT-BASED METHODS. *: HYBRID ARCHITECTURE-BASED METHODS)

Method	OA±STD(%)	
	20% Training	50% Training
GoogLeNet [40] †	83.44±0.40	86.39±0.55
VGGNet16 [41] †	86.59±0.29	89.64±0.36
D-CNN [9] †	90.82±0.16	96.89±0.10
ResNet50 [19] †	92.57±0.17	95.96±0.17
N-LDL-r [6] †	94.11±0.22	-
CSCANet [42] †	94.67±0.20	96.83±0.14
EfficientNet-B3 [43] †	94.79±0.19	97.21±0.16
Efficientnet-B3-Attn-2 [52] †	95.37±0.41	96.65±0.12
MGSNet [44] †	95.46±0.21	97.18±0.16
CGINet [24] †	95.35±0.14	97.10±0.24
ViT-B [21] ‡	94.80±0.46	96.66±0.46
V16_21k [45] ‡	94.97±0.01	-
MiniViT [46] ‡	95.43±0.17	97.05±0.12
SCViT [25] ‡	95.56±0.17	96.98±0.16
ET-GSNet [4] ‡	95.58±0.18	96.88±0.19
EMTCAL [43] *	94.69±0.14	96.41±0.23
Xu's Method [53] *	95.09±0.15	97.31±0.23
LSDGNet (ViT-B) ‡	95.68±0.30	97.39±0.18
LSDGNet (EfficientNet-B3) †	**95.81±0.14**	**97.49±0.19**

TABLE IV
EXPERIMENTAL RESULTS OF DIFFERENT METHODS ON NWPU DATASET. (†: CNN-BASED METHODS. ‡: VIT-BASED METHODS. *: HYBRID ARCHITECTURE-BASED METHODS)

Method	OA±STD(%)	
	10% Training	20% Training
GoogLeNet [40] †	76.19±0.38	78.48±0.26
VGGNet16 [41] †	76.47±0.18	79.79±0.15
D-CNN [9] †	89.22±0.50	91.89±0.22
ResNet50 [19] †	88.48±0.21	91.86±0.19
N-LDL-r [6] †	89.80±0.19	-
CSCANet [42] †	91.27±0.11	93.72±0.10
EfficientNet-B3 [43] †	92.91±0.12	94.96±0.09
MGSNet [44] †	92.40±0.16	94.57±0.12
CGINet [24] †	92.28±0.17	94.38±0.13
ViT-B [21] ‡	91.63±0.63	93.97±0.14
V16_21k [45] ‡	92.60±0.10	-
MiniViT [46] ‡	92.88±0.18	94.95±0.24
SCViT [25] ‡	92.72±0.04	94.66±0.10
ET-GSNet [4] ‡	92.72±0.28	94.50±0.18
EMTCAL [43] *	91.63±0.19	93.65±0.12
LSRL [54] *	93.45±0.16	94.27±0.44
Xu's Method [53] *	92.07±0.25	94.39±0.25
LSDGNet (ViT-B) ‡	92.75±0.30	94.67±0.10
LSDGNet (EfficientNet-B3) †	**93.60±0.17**	**95.05±0.11**

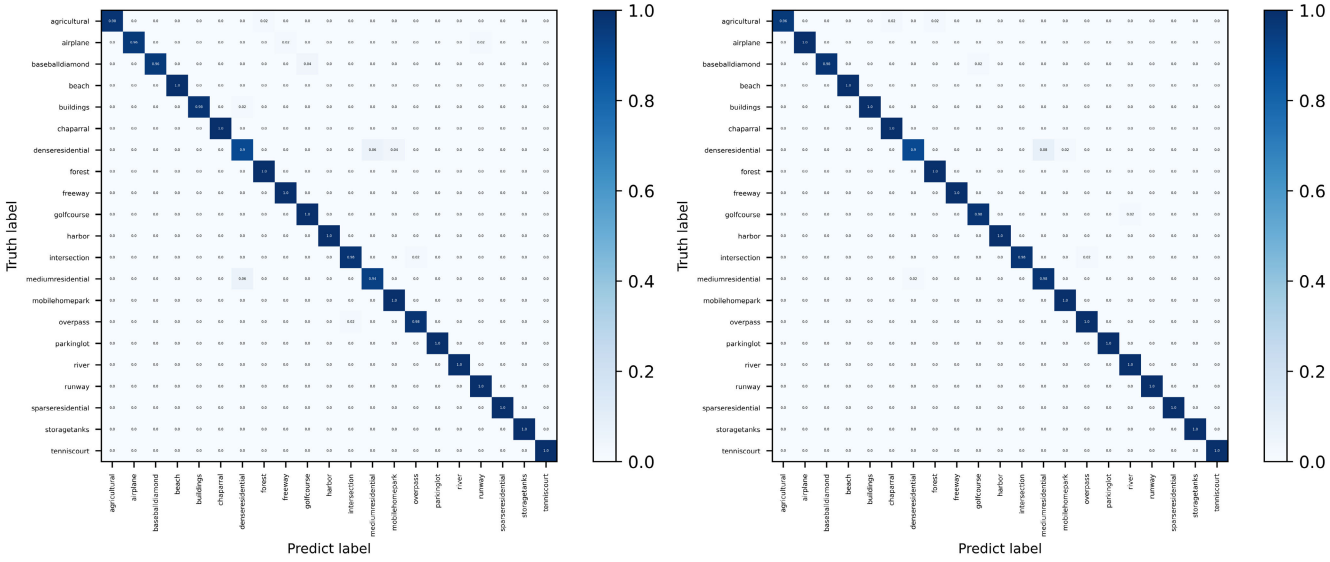
EfficientNet-B3 and LSDGNet (EfficientNet-B3) on the UCM dataset with 50% training data, respectively. From the figure, it can be seen that LSDGNet (EfficientNet-B3) has achieved improved classification accuracy in multiple categories, such as *medium residential*, *airplane*, and *buildings*. The above results demonstrate the effectiveness of our algorithm.

2) WHU-RS19 Dataset: The results on the WHU-RS19 dataset are shown in Table II. Among the compared methods, we achieve the highest accuracy. Specifically, when using 20% and 50% of the training samples, LSDGNet (ViT-B) achieves absolute improvements of 0.90% and 0.84%, respectively, compared with the original baseline model. LSDGNet (EfficientNet-B3) achieves absolute improvements of 2.53% and 0.83%, respectively. This demonstrates the effectiveness of our method on small-scale datasets. Fig. 7(a) and (b) shows the confusion matrices of EfficientNet-B3 and LSDGNet (EfficientNet-B3) on the WHU-RS19 dataset with 20% training data, respectively. From the figure, it can be seen that LSDGNet (EfficientNet-B3) has achieved improved classification accuracy in multiple categories, such as *beach*, *bridge*, *farmland*, and *industrial*. The above results demonstrate the effectiveness of our algorithm.

3) AID Dataset: The results on the AID dataset are shown in Table III. Among the compared methods, we achieve the highest accuracy. Specifically, LSDGNet (ViT-B) achieves an absolute improvement of 0.88% and 0.73% compared with the original baseline when using 20% and 50% of the training samples, respectively. Compared with SCViT, we achieve 0.12% and 0.41% improvement with 10% and 20% of training data, respectively. LSDGNet (EfficientNet-B3) achieves an absolute improvement of 1.02% and 0.28%, respectively. This demonstrates the effectiveness of our method on medium-sized datasets. Even on data with relatively few categories. Fig. 8(a) and (b) shows the confusion matrix of EfficientNet-B3 and LSDGNet (EfficientNet-B3) on the AID dataset with 20% training data, respectively. From the figure, it can be seen that LSDGNet (EfficientNet-B3) achieves different degrees of improvement in classification accuracy from 4% to 9% on *church*, *resort*, and *school*.

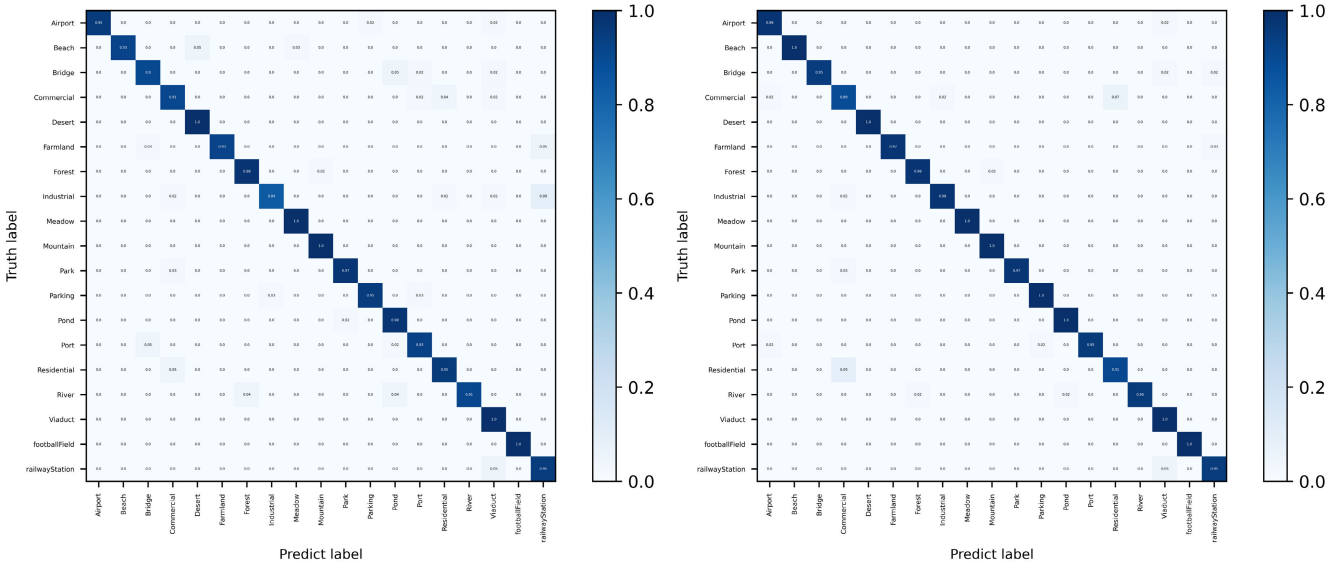
4) NWPU Dataset: The results on the NWPU dataset are shown in Table IV. Among the compared methods, we achieve the highest accuracy. Specifically, LSDGNet (ViT-B) achieves an absolute improvement of 1.12% and 0.7% compared to the original baseline when using 10% and 20% of training samples, respectively. Compared with

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(a) (b)

Fig. 6. Confusion matrix of EfficientNet-B3 and LSDGNet (EfficientNet-B3) on the UCM dataset with 50% training data. (a) Confusion matrix of EfficientNet-B3. (b) Confusion matrix of LSDGNet (EfficientNet-B3).



(a) (b)

Fig. 7. Confusion matrix of EfficientNet-B3 and LSDGNet (EfficientNet-B3) on the WHU-RS19 dataset with 20% training data. (a) Confusion matrix of EfficientNet-B3. (b) Confusion matrix of LSDGNet (EfficientNet-B3).

TABLE V
VALIDATION OF THE VALIDITY OF LSDGNET COMPONENTS

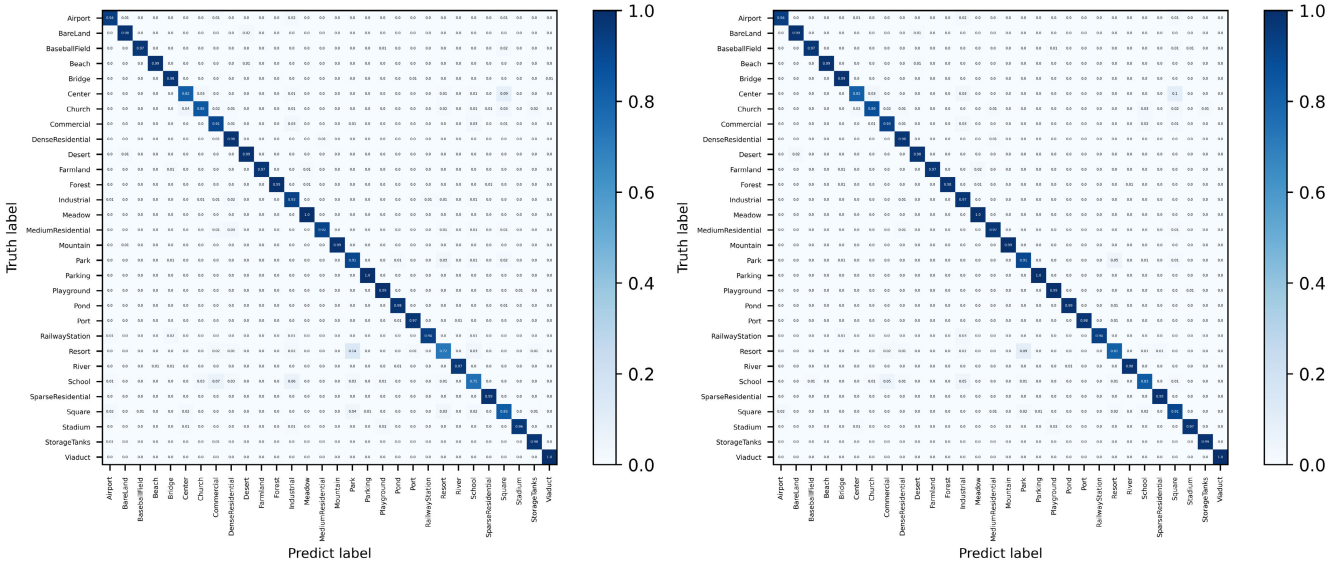
Base Method	DSLM	CFCM	NWPU 10% training
ViT-B			91.63
	✓		91.87
		✓	92.28
	✓	✓	**92.75**
EfficientNet-B3			92.91
	✓		93.15
		✓	93.26
	✓	✓	**93.60**

TABLE VI
COMPARISON BETWEEN USING ONLY COARSE-GRAINED CONTRAST LOSS OR USING ONLY FINE-GRAINED CONTRAST LOSS IN CFCM

Base Method	Mode	NWPU 10% training
ViT-B	with CFCM	92.75
	only with $\mathcal{L}_{Fine}$	91.95
	only with $\mathcal{L}_{Coarse}$	92.42
	none	91.87
EfficientNet-B3	with CFCM	93.60
	only with $\mathcal{L}_{Fine}$	93.27
	only with $\mathcal{L}_{Coarse}$	93.57
	none	93.15

other ViT-based methods, we achieve comparable accuracy. LSDGNet (EfficientNet-B3) achieves absolute improvements of 0.69% and 0.09%, respectively. This demonstrates the effectiveness of our method on large-scale multicategory datasets.

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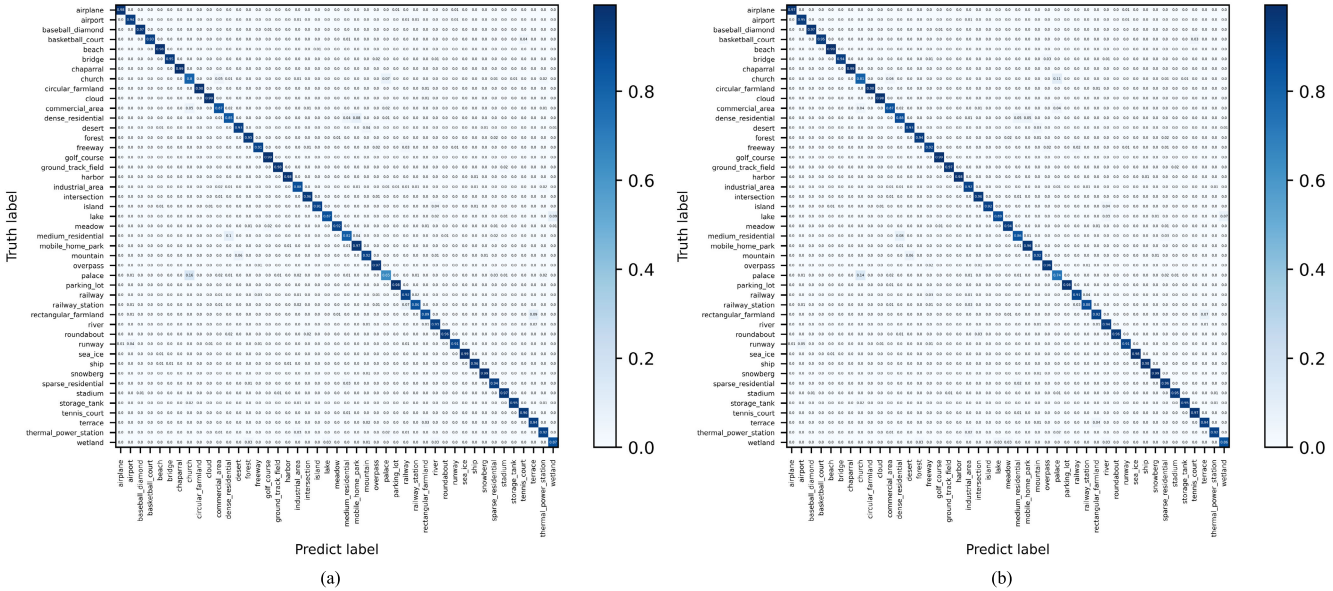


(a) (b)

Fig. 8. Confusion matrix of EfficientNet-B3 and LSDGNet (EfficientNet-B3) on the AID dataset with 20% training data. (a) Confusion matrix of EfficientNet-B3. (b) Confusion matrix of LSDGNet (EfficientNet-B3).

TABLE VII
PARAMETERS, FLOPS, AND INFERENCE TIME OF OUR METHOD

Method	Train		Test		Inference Time	NWPU(10%)
	Parameters	FLOPs	Parameters	FLOPs		
ViT-B	85.68M	22.00G	85.68M	22.00G	4.28ms	91.63
LSDGNet(ViT-B)	86.27M	22.03G	85.68M	22.00G	4.28ms	92.75
EfficientNet-B3	10.70M	1.32G	10.70M	1.32G	10.51ms	92.91
LSDGNet(EfficientNet-B3)	11.29M	1.35G	10.70M	1.32G	10.51ms	93.60



(a) (b)

Fig. 9. Confusion matrix of EfficientNet-B3 and LSDGNet (EfficientNet-B3) on the NWPU dataset with 10% training data. (a) Confusion matrix of EfficientNet-B3. (b) Confusion matrix of LSDGNet (EfficientNet-B3).

Because there are more difficult-to-distinguish fine-grained categories in large-scale multicategory datasets, our method effectively improves this problem by dynamically guiding the network to learn interclass relationships. Fig. 9(a) and (b) shows the confusion matrix of EfficientNet-B3 and LSDGNet (EfficientNet-B3) on the NWPU dataset with 10% training data, respectively. From the figure, it can be seen that LSDGNet (EfficientNet-B3) achieves classification accuracy improvement in several categories, such as *church*, *dense residential*, *medium residential*, and *palace*, where the

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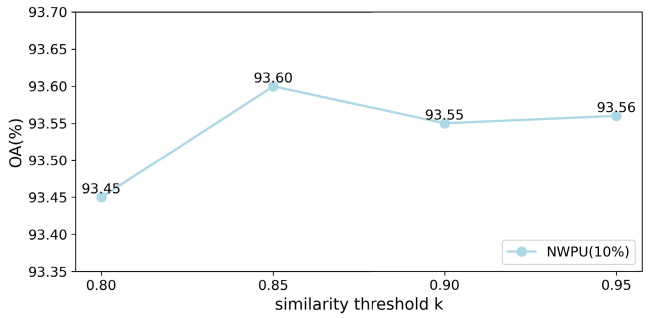


Fig. 10. Validation of the effect of different similarity thresholds k in label dynamic clustering on experimental results based on NWPU dataset (10% training data).

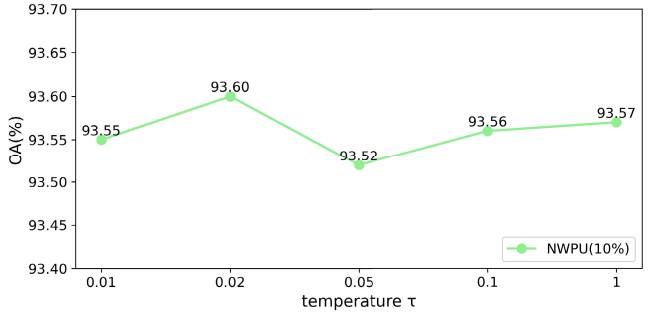


Fig. 11. Validation of the effect of temperature coefficient τ in DSLM based on NWPU dataset (10% training data).

confusion-prone category, *palace*, achieves classification accuracy from 65% to 74% absolute improvement, showing the effectiveness of our algorithm.

D. Ablation Study

We perform detailed ablation experiments in this section. All experiments are performed on the NWPU dataset with a training rate of 10%.

1) Effectiveness of the Components: In order to verify the effectiveness of each component of the proposed LSDGNet, ablation experiments are performed on each component. Specifically, we select ViT-B and EfficientNet-B3 as baseline on the NWPU dataset with 10% training data and add each component separately. The quantitative comparison is shown in Table V. We can see that both models under the LSDGNet bootstrap outperform the original baseline, which shows the effectiveness of LSDGNet.

2) Ablation Experiments With Similarity Threshold k: Verify the effect of different similarity thresholds k in label dynamic clustering on the experimental results. As shown in Fig. 10, we conducted experiments under the NWPU dataset (10% training data). From the experimental results, it can be seen that choosing a similarity threshold of 0.85 is optimal. This is because if the threshold is chosen too low, dissimilar categories will also be classified as the same coarse class, thus guiding the network to learn the wrong interclass relationship. Whereas if the threshold is chosen too high, there will be fewer fine-grained categories under the same coarse category, thus reducing the category association information.

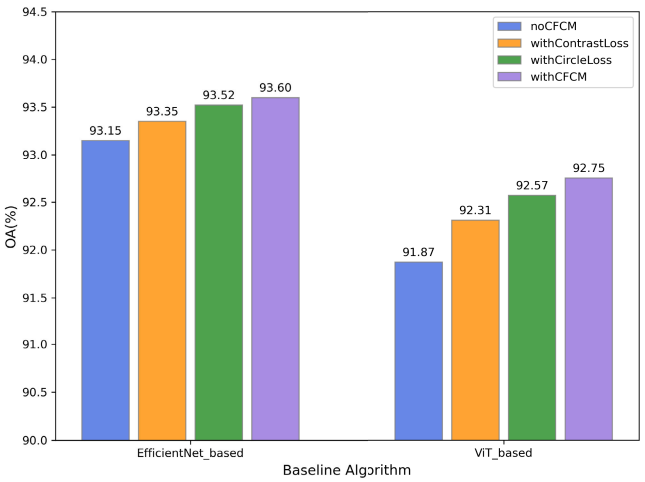


Fig. 12. Comparison based on NWPU (10% training data) using different contrast loss functions in CFCM.

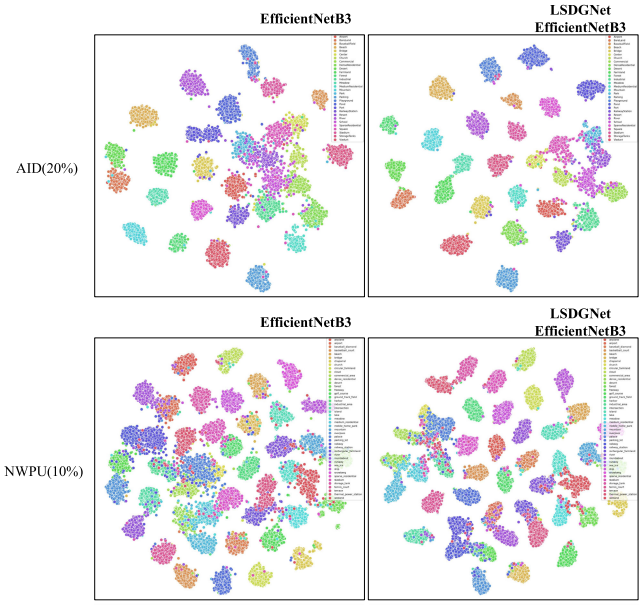


Fig. 13. t-SNE visualization results for EfficientNet-B3 and LSDGNet (EfficientNet-B3).

3) Ablation Experiment With Temperature Coefficient τ: The effect of temperature coefficient τ in the coarse intraclass softmax in the DSLM on the experimental results is shown in Fig. 11. As τ increases, the experimental results show a trend of increasing and then decreasing. This is because when τ is small, it causes the distribution of the generated interclass relations to be close to that of hard labels. Also, a larger τ will reduce the gap of interclass similarity.

4) Ablation Experiments With Different Loss Functions: We compare the effect of the CFCM with other contrast losses. As shown in Fig. 12, we compare the effects of ContrastLoss [55], CircleLoss, and CFCM, respectively. From the experimental results, it can be seen that the results obtained from the coarse-to-fine decoupled contrast loss are higher than the other losses on both base algorithms. This proves the effectiveness of our algorithms. In addition, we conduct ablation

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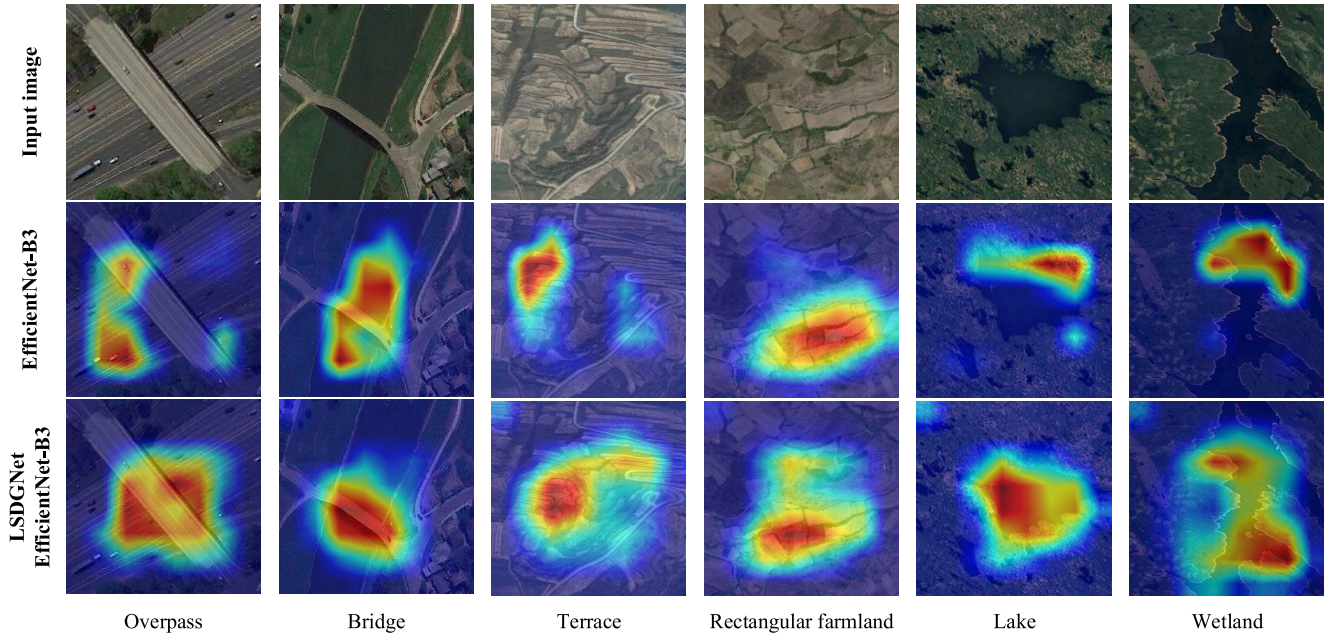


Fig. 14. Grad-CAM++ visualization results for EfficientNet-B3 and LSDGNet (EfficientNet-B3).

experiments on coarse-grained loss and fine-grained loss in the CFCM module. As shown in Table VI, when only fine-grained loss is used, the OA based on ViT-B and EfficientNet-B3 methods decreases by 0.80% and 0.33%, respectively. When only using coarse-grained loss, the OA based on ViT-B and EfficientNet-B3 methods decreases by 0.33% and 0.03%, respectively. This is because using only coarse-grained loss or fine-grained loss lacks effective guidance on the distance between feature vectors.

E. Parameter Quantity and Computational Complexity

Table VII shows the parameter count, computational complexity, and inference time of our algorithm. Compared with the baseline model, LSDGNet (ViT-B) and LSDGNet (EfficientNet-B3) require fine-tuning of the linear layers during the training phase, which introduces a small amount of additional parameters and computational complexity. In the model inference stage, the cross-modal guidance branch can be completely removed, so no additional parameters and computational complexity will be introduced to the original model.

F. Visualization

The feature distribution of different categories of images can reflect the model's ability to learn about intraclass and interclass relationships. t-SNE [13] is a nonlinear feature dimensionality reduction method that maps high-dimensional features into a low-dimensional space to visualize feature distribution information. We visualize the classification results using t-SNE on two datasets, NWPU (10% training data) and AID (20% training data), respectively. As shown in Fig. 13, the feature vectors encoded by LSDGNet (EfficientNet-B3) have more pronounced intraclass tightness and interclass discretization compared to EfficientNet-B3. This indicates that our method can better guide the network to learn intraclass and interclass relationships.

In addition, in classification tasks, discriminative features are highly correlated with semantic labels, so locating salient regions of the image that the model focuses on provides a more intuitive representation of the model's feature extraction capabilities. Grad-CAM++ [56] is a popular visualization method that intuitively demonstrates the regions of the image that the model focuses on. Fig. 14 shows the visualization results of EfficientNet-B3 with LSDGNet (EfficientNet-B3). Both *overpass* and *bridge* contain bridges. LSDGNet (EfficientNet-B3) classifies *overpass* and *bridge* by locating bridge buildings and their surrounding backgrounds. From a distant perspective, *terrace* and *rectangular farmland* have similar color and texture features. LSDGNet (EfficientNet-B3) distinguishes them by focusing on a wide range of image information. There is an area with water in both *lake* and *wetland*. The difference lies in whether the area is enclosed. LSDGNet (EfficientNet-B3) focuses on the overall water area to distinguish between these two categories. The above example can demonstrate the effectiveness of our proposed algorithm for classification discriminative feature extraction.

V. Conclusion

In this article, we introduce semantic information from labels into the task of remote sensing image scene classification. We design a novel architecture that leverages this semantic information to dynamically guide the model in learning intraclass and interclass relationships at both the target and feature levels. Specifically, we encode the label text using a LLM and a text encoder, clustering the semantic

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representation vectors into multiple coarse-grained categories using a hierarchical clustering algorithm. Based on the outcomes of this semantic dynamic clustering, we develop a DSLM and a CFCM. The DSLM facilitates the network's understanding of relationships between different categories at the target level through soft labels, while the CFCM enhances feature-level learning of class relationships using contrastive learning principles. We validate our approach by selecting two popular classification algorithms based on CNN and ViT as baselines, conducting experiments on four benchmark datasets, and demonstrating the effectiveness of our methods through a series of ablation and visualization experiments.

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